CSE 150A Final Project

Cozy Markov Blankets

1 Problem Description

We want to address whether sleep quality can be predicted each day by various observable factors, including total phone usage, night-time phone usage, awake percentage, sleep onset latency, and actual sleep duration. This problem is important because high sleep quality is essential for maintaining cognitive and physical health. Poor sleep can affect our attention, memory, mood, growth hormones, and immune system. By identifying the daily behaviors that have strong predictive value for sleep quality, we can discover new habits and routines that support a good night's sleep. In relation to probabilistic reasoning, sleep quality is a subjective measure that can be influenced by numerous factors at once. Creating a model will help us represent sleep quality using quantitative features we are able to measure, as well as adjust to the variability that comes with observing human behavior.

2 Data Sourcing and Processing

We will be using the SleepQual and B.Health data set (Arora et al., 2022) from Kaggle. The dataset contains 7 consecutive days of sleep patterns from 24 university students, totaling to 182 rows, 51 columns, and no missing data. The data set contains many variables that are relevant to our problem, in which our model can utilize to make better inferences about the hidden sleep quality state. However, one limitation of the dataset is that many columns are repetitive because they convey the same observations but take on different formats (continuous, percentages, discrete). To work around this limitation, we will only be using the continuous columns of the observations (i.e. those measured in minutes). In addition, the data set provides a large number of variables. To avoid the curse of dimensionality/sparse data, we will be using five variables to guarantee model stability, listed in the table below.

Table 1: Features for model inference

Variable name	Definition
`total phone usage / day (minutes)`	total phone usage in minutes
`night time phone usage / day (minutes)`	phone usage in minutes during the night
`awake percentage (%)`	percentage of time in bed spent awake
`sleep onset latency (minutes)`	time it takes to fall asleep in minutes
`actual sleep duration (minutes)`	actual sleep duration in minutes

We will be normalizing all features before inputting them to the model using `sklearn.preprocessing.StandardScaler`. This will ensure variables with larger values do not dominate variables with smaller values, and thus our model will perform more efficiently and accurately.

3 Modeling and Inference

For the problem, we will be doing inference with the Viterbi algorithm, with the data represented as a Hidden Markov Model. Sleep quality is suitable with a HMM because sleep quality isn't something that is completely observable, and is subjective dependent on each person. In addition, sleep quality

from one night to another is sequential. That is, sleep quality from a previous night can impact sleep quality on a subsequent night. To simplify, we will be assuming sleep quality is only affected by one prior night, and only has an effect on sleep quality for one subsequent night.

Our hidden state of sleep quality for night t is represented by the ternary variable S_t, with values of 0 (good sleep quality), 1 (moderate sleep quality), and 2 (bad sleep quality). Each state S_t will have 5 features corresponding to it: total phone usage, night-time phone usage, awake percentage, sleep onset latency, and actual sleep duration. A given observation O_t will be the set of those 5 features. The initial state distribution will represent the sleep quality of the person on the first night, without prior knowledge of the person. This will be a uniform distribution. The transition probabilities will be a matrix where the (i, j) entries represent the probability of going from the ith sleep quality state to the jth sleep quality state. These transition probabilities are non-deterministic because a person can have two good sleeps in a row, two bad sleeps in a row, a good sleep followed by a bad sleep, a bad sleep followed by a good sleep, and so on and so forth.

Typically, the probability of emissions would be represented as a matrix. However, since our observations will be feature vectors that contain continuous data, we will be using Gaussian distributions to represent each state. We will be using the Baum-Welch (forward-backward) algorithm to learn the probabilities for the transition matrices and the parameters for the Gaussian emission distribution. We will be using the dataset mentioned above for our observations.

4 Results and Discussion

4.1 Model Configuration & Learned Parameters (Initial Model)

Our HMM was implemented using the `hmmlearn.hmm.GaussianHMM` class library. The specific model type Gaussian HMM was chosen because our dataset contained five continuous numerical features as observed variables: total phone usage per day (minutes), night time phone usage per day (minutes), awake percentage, sleep onset latency (minutes) and actual sleep duration (minutes).

The model architecture consisted of three hidden states corresponding to sleep quality levels derived from SleepQual scores: Good (≤ 3), Fair ($4-6$), and Poor (≥ 7). We chose not to utilize a higher number of hidden states for our initial model to make results more interpretable for future revision. To train our model we utilized the Baum-Welch algorithm (`model.fit(X)`) to learn the three main parameters of the HMM: initial state probabilities, the transition matrix, and the Gaussian emission parameters. Then using these parameters the Viterbi algorithm was implemented to infer the most probable sequence of hidden states given the observations.

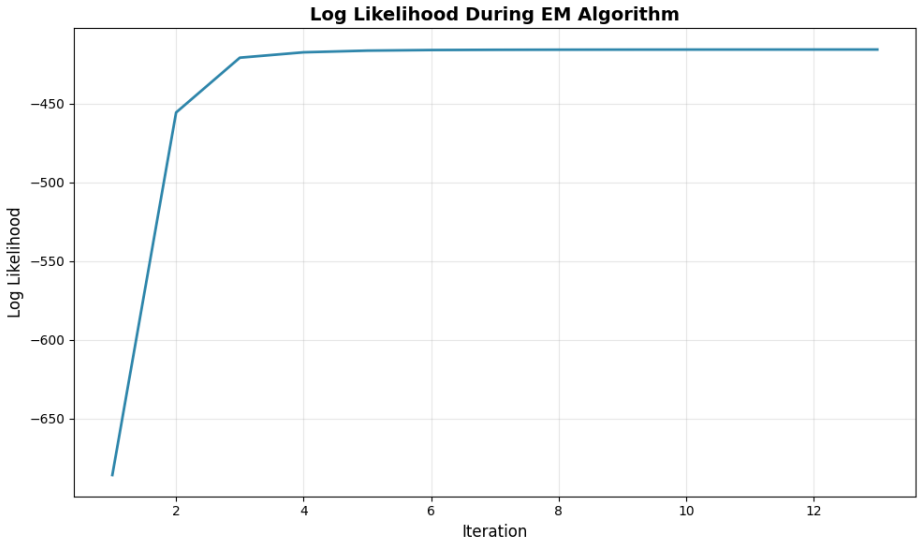


Figure 1: Log likelihood of EM algorithm in the Hidden Markov Model.

4.2 Quantitative Performance

Our model's performance on predicting the known sleep quality labels were evaluated based on two metrics:

Viterbi Reconstruction Accuracy: 0.7802

- This metric measures the percentage of observations where the decoded hidden state from our algorithm matched the corresponding ground truth label. Our accuracy seems to be relatively high, implying that the model is not struggling to map the decoded states to the ground truth states. The model successfully predict 142 out of 182 observations correctly.

Mean Absolute Error: 0.2253

- This metric measures the average absolute difference between the ground truth label and the decoded label. As our labels are 0 (Good), 1 (Moderate), and 2 (Poor), an MAE of 0.2253 implies that the model is predicting sleep quality labels within approximately one-quarter of a level on average.

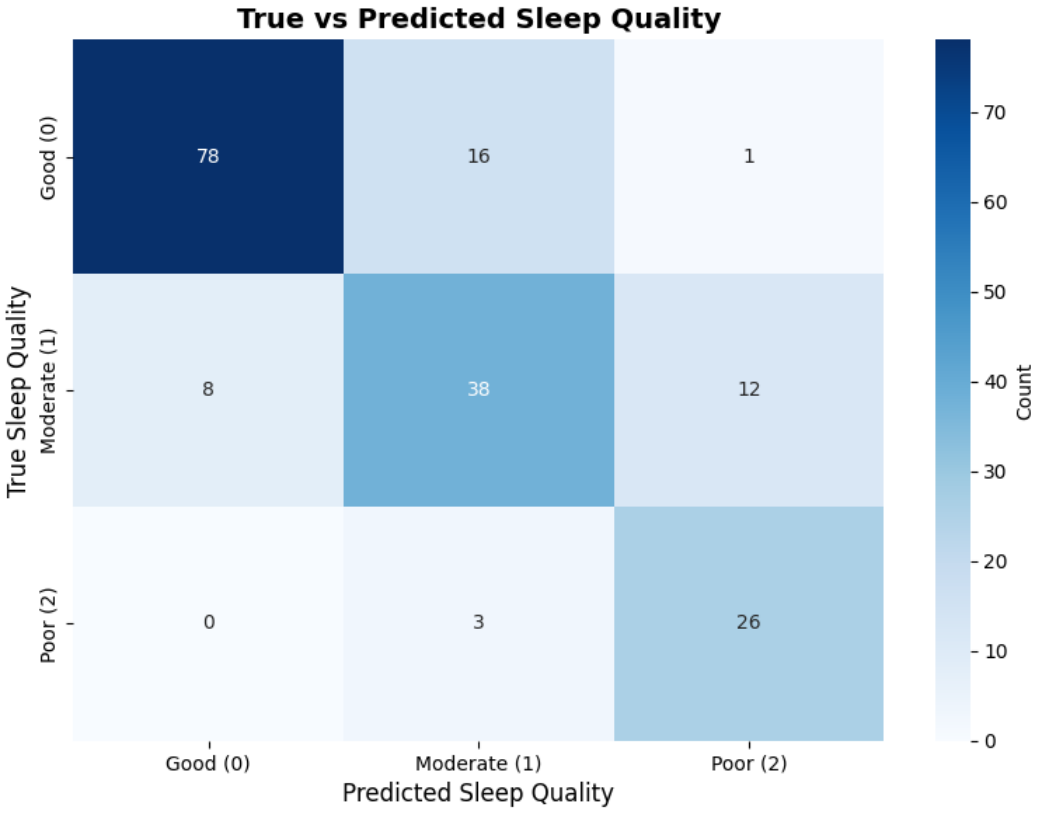


Figure 2: Heat map of EM model results.

The model demonstrates strong performance in identifying the extreme categories of Good and Poor sleep quality, with slightly more error in the Moderate category.

Furthermore, when testing for consistency across the number of iterations, MAE decreases, as expected, however, performance seems to drop off.

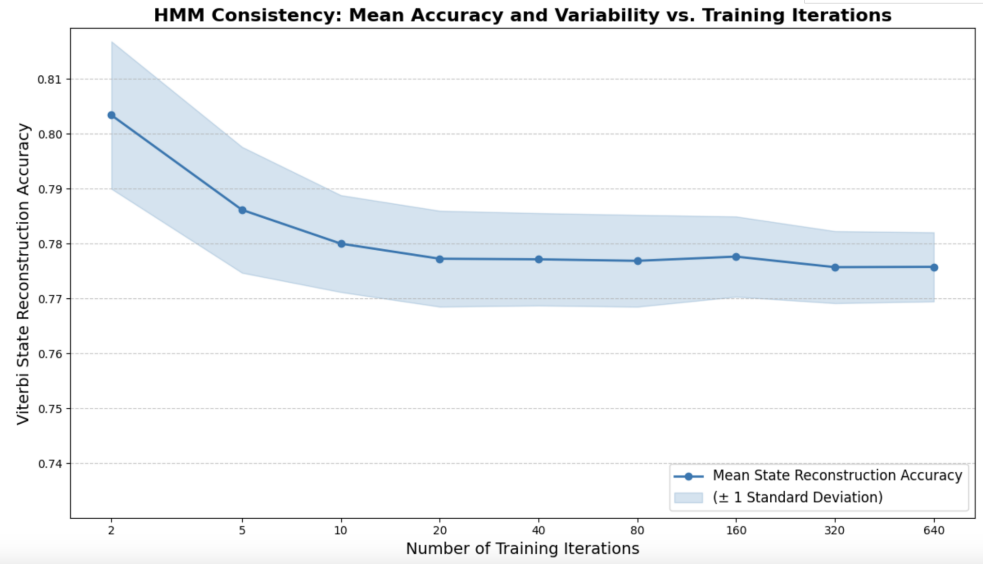


Figure 3: Accuracy of model across EM iterations.

4.3 Qualitative Insights & Discussion

Based on the model results, the model successfully captures the extremes of sleep quality, demonstrating strong performance in distinguishing between Good and Poor sleep states. This suggests that the observed features (phone usage, sleep duration, and sleep onset latency) provide clear distinguishing factors for extreme sleep conditions. However, the model shows greater uncertainty when predicting Moderate sleep quality. This uncertainty arises because moderate sleep occupies a transitional range, where indicators of Good and Poor sleep overlap, making it challenging to accurately predict sleep quality based on the five observable features.

Our model's learned distributions seems to align with intuition. For example, the model makes a distinction between observations with high, medium, and low awake percentage (Figure 4). Observations with high awake percentages seem to have Poor sleep states, while observations with low awake percentages tend to have Good sleep states.

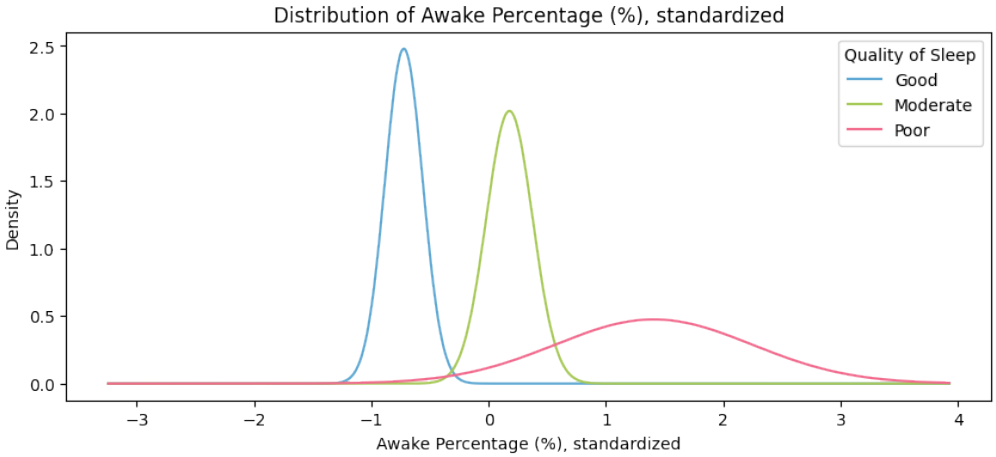


Figure 4: Learned distribution of the *awake percentage (%)* feature through EM.

4.4 Convergence & Scalability

The training demonstrated stable convergence, with the model reaching its optimal log-likelihood at around 18 iterations out of a maximum 200. This monotonic improvement in log-likelihood throughout the training confirms that our model was properly implemented.

Additional data would likely improve this model's performance, especially for the underrepresented Poor sleep category. The computational cost would remain minimal since our current model runs efficiently on 182 observations.

5 Conclusion

5.1 Summary of Key Findings and Model Performance

Our Gaussian HMM achieved a Viterbi reconstruction accuracy of 76.92% and a MAE of 0.225 in predicting sleep quality from various observable factors. The model effectively captured extreme sleep states, correctly categorizing 82% of Good sleep instances and 90% of Poor sleep instances, while showing uncertainty in the Moderate category. Training converged efficiently after 18 iterations, indicating stable parameter learning. The five observable factors (total hone usage, night-time phone usage, awake percentage, sleep onset latency, and actual sleep duration) provided sufficient predictive capability to distinguish sleep quality states. These results depict that an HMM framework can successfully model the relationship between daily behavioral habits and sleep quality outcomes.

5.2 Limitations and Potential Improvements

Limitations

Internal State Classification:

- The learned HMM states may not correspond to the ground truth HMM states. For instance, if the model decides to use a 0 as "high quality" sleep and 2 as "low quality," it does not match up with the dataset and may throw the entire model off without additional investigation.

Gaussian Emission Distribution:

- Using a Gaussian distribution for all is a strong assumption that may not hold over for larger datasets, especially datasets with discrete variables of interest.

Markov Order:

- The assumptions made by a Markov process may be too simplistic to fully encapsulate complex human behavior, which could be affected by longer-term issues not in the data for the observations.

Improvements

Adjust Number of Hidden States:

- Reducing to 2 hidden states simplifies the model and may improve alignment with the data clusters, while increasing to 4-5 hidden states can capture finer distinctions between sleep quality. However, more hidden states would be computationally expensive and would require more training samples.

Adjust Emission Distribution:

- Instead of assuming a single Gaussian distribution for each hidden state, the model could use more flexible emission distributions that better capture multimodal or skewed patterns in the data. This would allow the HMM to represent more realistic observation behaviors and reduce bias. However, these models would also increase computational cost.

6 Reflections & Contributions

6.1 Advice for Future Students

For future students trying out this project, we suggest finding a larger dataset with day to day variability because HMMs are naturally limited by the data they are given. It is also helpful to limit the number of prediction values that the model can produce, not only to improve the metrics, but also to simplify the complexity of the project. Exploratory data analysis is very important to highlight patterns and potential issues within the data, so graphs are especially beneficial. For modeling and evaluation, start with a simple HMM and use many metrics, rather than only accuracy, to determine the predictive proficiency of the model.

6.2 Team Members

[Team Member]: I contributed to the project by helping create visualizations, developing the initial model, and contributing to the write-up. From this project, I learned how to implement an HMM using real-world sleep and behavioral data, strengthening my understanding of how probabilistic frameworks can uncover latent patterns in sequential data. I also gained valuable practical experience applying the Baum-Welch and Viterbi algorithms to parameter learning and state inference.

[Team Member]: I contributed to the project by helping with data cleaning, initial modeling, and the writeup. Taking the time to find a good dataset is really important for a project like this. Originally, we were using a dataset that was not suitable for our model, and had to spend a long time figuring out why our metrics were poor. I understand Hidden Markov Models better after completing this project. I additionally learned the importance of consistent communication and efforts with my team.

[Team Member]: I contributed to the project by helping with initial EDA, model revisions, visualizations, and the writeup. From this project I got more practice in creating visualizations and familiar with how the probability models work under the hood and react under certain parameters. In terms of the project, I definitely think communicating with your group members and understanding the requirements are crucial.

[Team Member]: I contributed to the project by helping with improving the model, visualizations, and the writeup. This project was really insightful to my practical understanding of Hidden Markov Models. After learning the Viterbi and forward-backward algorithms in class with little intuition, I finally understand both algorithms and how it fits into the HMM pipeline. In addition, I understand why the HMM has a sequential aspect to it and what can happen when that criteria is not met.

6.3 Generative AI

We used generative AI to help us better understand the theory behind Hidden Markov Models and to suggest model improvement strategies when our performance was low.

7 References

[1] Anshika1011. (2022) SleepQual and B.Health Dataset. In *Kaggle Datasets*. Retrieved from https://www.kaggle.com/datasets/anshika1011/sleepqual-and-bhealth-dataset/.

[2] Arora, A., Chakraborty, P. & Bhatia, M.P.S. (2021) Intervention of Wearables and Smartphones in Real Time Monitoring of Sleep and Behavioral Health: An Assessment Using Adaptive Neuro-Fuzzy Technique. *Arabian Journal for Science and Engineering*, pp. 1-26.

[3] Arora, A., Chakraborty, P. & Bhatia, M.P.S. (2022) SleepQual and B. Health: Smartwatch and Smartphone based Behavioral Datasets of Youth. In 2022 12th International Conference on Cloud Computing, Data Science & Engineering. *Confluence*, pp. 340-344. IEEE.